

ARK-448 Preregistration — Dynamical Decoupling vs. Baseline Under an Idle Window

Status:  LOCKED (preregistered before execution)

Experiment ID: ARK-448

Series: ExecutionProof ARK (Authorization-boundary Repeatability Kit)

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Protocol: Field 27 (LOCK → SPAM gate → principal job → analyze → verdict)

1. Motivation

ARK-447 compared **Pauli Twirling vs. Baseline** on the single-round authorization boundary and explicitly **omitted Dynamical Decoupling (DD)** due to circuit-scheduling complexity. ARK-448 closes that gap: it is the deferred DD experiment.

DD suppresses **dephasing (T2)** noise that accumulates while a qubit sits idle. Real authorization workflows are not instantaneous — a decision qubit may hold its state across an idle window before the

boundary gate fires. ARK-448 asks a concrete, falsifiable question:

When an idle window is deliberately inserted before the authorization boundary gate, does an XX dynamical-decoupling sequence during that window improve the authorization boundary (higher ALLOW retention S_A and/or lower DENY leakage L_D) relative to a bare idle delay?

2. Hypothesis (stated before execution)

- **H1 (DD helps):** DD arms show higher S_A and/or lower L_D raw than baseline arms.
- **H0 (null):** DD produces no statistically significant change (Δ within noise), OR DD is net negative because the extra X-pulse gate overhead adds more error than the dephasing it removes.

We explicitly pre-commit to reporting whichever outcome occurs, including a null or negative

result. The dominant decay channel of the ALLOW authorizer state $|1\rangle$ during idle is **amplitude damping (T1)**, which DD does not mitigate; DD principally protects the superposition (T2). We therefore consider a null / modest / negative DD effect a genuinely plausible and scientifically valid outcome, consistent with the honest-reporting ethos demonstrated by ARK-445 (FAIL).

3. Design

Four arms, submitted in a **single principal job** on one backend and one qubit pair, so DD-on vs DD-off is compared with no cross-job confound. DD is **baked into the circuits** (explicit delays +

X pulses), not applied as a runtime option, ensuring baseline and DD arms differ only by the pulse insertion.

Arm	Authorizer (Q_A)	Idle window	DD sequence
arm1_ALLOW_baseline	$ 1\rangle$ (X)	τ bare delay	none
arm2_DENY_baseline	$ 0\rangle$	τ bare delay	none
arm3_ALLOW_dd	$ 1\rangle$ (X)	τ	XX
arm4_DENY_dd	$ 0\rangle$	τ	XX

- **Payload (Q_P):** prepared in $|+\rangle$ (H), as in ARK-447.
- **Boundary gate:** CNOT(Q_A $\rightarrow$ Q_P) after the idle window.
- **Idle window:** $\tau = 20\ \mu\text{s}$, applied to BOTH Q_A and Q_P after state preparation and before the boundary CNOT.
- **DD sequence (XX):** `delay( $\tau/4$ ) – X – delay( $\tau/2$ ) – X – delay( $\tau/4$ )` on both qubits (net idle = τ , two X pulses). This is the standard two-pulse CPMG-style refocusing sequence.
- **Measurement & analysis convention:** identical to ARK-447 (measure Q_A $\rightarrow$ c[0], Q_P $\rightarrow$ c[1]; metrics computed on the same classical bit as ARK-447 for direct comparability).

4. Fixed parameters (LOCKED)

- **Backend:** `ibm_marrakesh` (series-consistent). Fallback `ibm_fez` only if marrakesh is non-operational at submission time; the backend actually used is recorded in the proofrecord.
- **Qubit selection:** lowest connected-pair readout-error sum (same selector as ARK-447).
- **Shots:** principal job **8192/circuit**; SPAM gate **2048/circuit**. If the post-SPAM budget check shows < 90 s remaining, principal shots drop to 4096 (recorded in RUN_LOG). No other design change.
- **Transpilation:** `optimization_level=1`, delays preserved (scheduling honored on hardware).

5. SPAM gate (must pass before principal job)

- SPAM_A: prepare $|1\rangle$ on Q_A, measure Z. Require readout error ≤ 0.02 .
- SPAM_P: prepare $|+\rangle$ on Q_P, measure Z. Require $|P('1') - 0.5| \leq 0.02$.
- **SPAM_P is a GATING diagnostic ONLY.** Per the ARK-447 v1.1 correction, the ~ 0.5 outcome is the expected physics of a superposition and is **NEVER subtracted** from DENY leakage. DENY leakage is reported RAW.

6. Metrics (per arm)

- **S_A** = fraction of ALLOW shots reading '1' on the authorizer classical bit.

- **L_D_raw** = fraction of DENY shots reading '1' on the same bit (RAW leakage, no SPAM subtraction).
- **Δ_B** (boundary margin) = $S_A - L_{D_raw} - 0.20$.

7. Pass criteria (per configuration)

- $S_A \geq 0.90$
- $L_{D_raw} \leq 0.02$
- $\Delta_B \geq 0.00$

A configuration PASSES only if all three hold.

8. Verdict rules (stated before execution)

- **PASS (strong):** both baseline and DD pass AND DD improves S_A or L_{D_raw} .
- **PASS (weak):** both pass but DD shows no improvement (null).
- **MIXED:** exactly one of {baseline, DD} passes.
- **FAIL:** neither passes (e.g., the 20 μ s idle window degrades the boundary below threshold — an honest negative demonstrating idle-time sensitivity).

Primary statistical comparison: two-proportion z-test on S_A (DD vs baseline) and on L_{D_raw} (DD vs baseline). Significance threshold $\alpha = 0.05$ (two-sided).

9. Interpretation boundaries

- Results apply only to this backend, qubit pair, calibration snapshot, and $\tau = 20 \mu$ s.
- This is a hardware noise-mitigation study, **not** a cryptographic security validation.
- DD is error mitigation, not error correction; no QEC is used.
- Single-round binary boundary only; not a multi-round or production protocol.
- The authorizer $|1\rangle$ state decays primarily via T1, which DD does not address; a limited DD benefit is expected and does not indicate methodological failure.

10. Files (to be committed at LOCK)

- `ARK_448_preregistration.md` (this file)
- `ark_448_select_qubits.py`
- `ark_448_spam_job.py`
- `ark_448_circuits.py`
- `ark_448_submit_ibm.py`
- `ark_448_retrieve.py`
- `ark_448_analysis.py`
- `MANIFEST.txt` (SHA-256 of all locked files)

Post-execution artifacts (`selected_qubits.json` , `spam_results.json` , `principal_job_*` , `raw_results.json` , `proofrecord.json` , `RESULTS.md` , `RUN_LOG.md` , `README.md`) are added after the run and never alter the locked preregistration.

LOCK statement: The hypothesis, arms, idle window, DD sequence, shot counts, SPAM gate, metrics, pass criteria, and verdict rules above are fixed prior to execution. Any deviation forced by hardware (e.g., shot reduction under budget) is logged in `RUN_LOG.md` and does not alter these criteria.